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SECTION A

Q1.

Role of Regularization in Machine Learning

Regularization mitigates overfitting by introducing a complexity penalty to the model's loss function. It constrains the magnitude of the learned coefficients to ensure the model captures underlying patterns rather than training noise. This process optimizes the bias variance trade-off to improve generalization on unseen data.

Q2.

Types of Regularization Techniques

- **L1 Regularization (Lasso):** This technique adds a penalty equivalent to the absolute value of the magnitude of coefficients. It frequently shrinks less important feature coefficients to zero, effectively performing feature selection.

Example: Gene expression analysis where thousands of genes are measured but only a small subset dictates the phenotype.

- **L2 Regularization (Ridge):** This method adds a penalty equivalent to the square of the magnitude of coefficients. It penalizes large weights but rarely reduces them to zero, ensuring that all features contribute to the model.

Example: Predicting credit solvency where multiple correlated financial factors all contribute marginally to the final risk score.

Q3.

High-Dimensionality with Limited Samples ($p \gg n$)

a. Algorithms to Avoid:

- **Unregularized Deep Neural Networks:** These require vast data to tune their massive parameter sets and will memorize the training set in low sample regimes.
- **K-Nearest Neighbors (KNN):** Distance metrics lose meaningfulness in high dimensional space because all data points become roughly equidistant.

b. Dimensionality Reduction Techniques:

- **Principal Component Analysis (PCA):** This linear technique is ideal for projecting data onto orthogonal axes that maximize variance while discarding noise.
- **UMAP (Uniform Manifold Approximation and Projection):** This non linear method preserves local neighborhood structure better than PCA and is superior for visualizing clusters in complex manifolds.

Q4.

Naïve Bayes Classification

a. **Derivation from Bayes Theorem:** The Bayes Theorem is defined as:

$$P(C|x) = \frac{P(x|C)P(C)}{P(x)}$$

Where:

- $P(C|x)$ is the posterior probability of class C given feature vector x .
- $P(x|C)$ is the likelihood of features given the class.
- $P(C)$ is the prior probability of the class.
- $P(x)$ is the evidence (marginal likelihood).

Bayes' Theorem for a class C_k and feature vector $\mathbf{x} = (x_1, \dots, x_n)$

$$P(C_k|x) = \frac{P(C_k) * P(x|C_k)}{P(x)}$$

Apply the "naive" assumption that all features x_i are mutually independent given the class C_k .

Therefore:

$$\begin{aligned} P(x|C_k) &= P(x_1|C_k) \times P(x_2|C_k) \times \dots \times P(x_n|C_k) \\ &= \prod_{i=1}^n P(x_i|C_k) \end{aligned}$$

Since the denominator $P(\mathbf{x})$ is constant across all classes, the classification rule becomes:

$$\hat{y} = \operatorname{argmax}_k \left(P(C_k) \prod_{i=1}^n P(x_i | C_k) \right)$$

b. Classification Process The algorithm computes the posterior probability for every possible class. It multiplies the prior probability of the class by the conditional likelihood of each feature value given that class. The system then assigns the input vector to the specific class that yields the highest calculated probability score.